

GRA-CellTwin-Final: A Multilevel GRA Zeroing Framework for Accurate In Silico Cell Twins and Personalised Therapy Generation

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Abstract

We present **GRA-CellTwin-Final**, a multi-level simulator that builds reliable *in silico* replicas (digital twins) of single cells using the GRA (Generalized Resonance Alignment) meta-zeroing framework. The system integrates public transcriptomic data (CellxGene, TCGA, GEO) with a “multiverse foam” of candidate cell states. By computing a critical foam threshold Γ_{crit} and a resonance frequency ω_{res} , the algorithm eliminates hallucinated or inconsistent states, guaranteeing ethically filtered, explainable outputs. GRA-CellTwin can generate personalised therapies including Yamanaka factors (OSKM), senolytics, and CRISPR targets. The open-source implementation includes a Jupyter notebook and integration with Ollama/LLM for post-processing. We demonstrate the pipeline on breast cancer data, producing a validated therapy proposal. This work provides a foundation for zero-hallucination digital cell twins, with potential applications in precision oncology, regenerative medicine, and drug discovery.

1 Introduction

Reliable *in silico* models of single cells are crucial for understanding disease mechanisms and designing personalised therapies. However, existing approaches often suffer from hallucinations, overfitting, or lack of interpretability. The GRA (Generalized Resonance Alignment) meta-zeroing framework offers a principled way to filter inconsistent states by modelling a “multiverse foam” of candidate hypotheses and nullifying those that deviate beyond a critical resonance band.

GRA-CellTwin-Final implements this framework specifically for single-cell transcriptomics. It:

- Loads public scRNA-seq datasets (GEO, TCGA, CellxGene).
- Builds a foam of hypothetical cell states (bubbles) around the observed transcriptome.
- Computes resonance frequencies and removes non-coherent bubbles (hallucinations).
- Generates actionable therapies: OSKM reprogramming, senolytics, CRISPR guides.
- Applies an ethics filter (“first, do no harm”) to block unsafe proposals.

This paper describes the mathematical formalism, software architecture, and a demonstration on breast cancer data.

2 Mathematical Formalism of GRA Zeroing for Cell States

2.1 Multiverse Foam Representation

Let a single cell be represented by its transcriptomic profile $\mathbf{x} \in \mathbb{R}^d$ (expression of d genes). We generate a population of N candidate states (“bubbles”) $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ by adding controlled noise to the observed profile or by sampling from a generative model. Each bubble also carries a weight w_i (initialised equally).

2.2 Foam Density and Critical Threshold

Define the local foam density around a bubble $\mathbf{x}_i$ as the average distance to its k nearest neighbours:

$$\rho_i = \frac{1}{k} \sum_{j \in \text{kNN}(i)} \|\mathbf{x}_i - \mathbf{x}_j\|.$$

The global critical foam threshold is set to:

$$\Gamma_{\text{crit}} = \min\{\min_i \rho_i, T_c\},$$

where T_c is a user-defined hyperparameter (e.g., 0.3). Bubbles with $\rho_i > \Gamma_{\text{crit}}$ are considered too isolated (potential hallucinations).

2.3 Resonance Frequency

We compute a spatial correlation matrix R_{ij} between bubbles. The resonance frequency of the multiverse is:

$$\omega_{\text{res}} = \sum_{i,j} \Gamma_{\text{crit}} R_{ij}.$$

Bubbles whose pairwise correlation falls outside the band $[\omega_{\text{res}} - \delta, \omega_{\text{res}} + \delta]$ are removed. This step eliminates inconsistent states that do not resonate with the majority.

2.4 GRA Zeroing and Ethics Filter

The GRA zeroing operator removes all bubbles that fail either the density or the resonance criterion. The remaining bubbles define a coherent multiverse of plausible cell states. An ethical filter then evaluates any proposed therapy against a rule-based system (e.g., disallowing therapies that cause predicted harm). Only therapies passing the filter are output.

3 System Architecture and Implementation

3.1 Data Loading and Preprocessing

The module `cell_models/cell_twin.py` loads scRNA-seq data from:

- GEO (direct download by accession),
- TCGA (e.g., BRCA),

- CellxGene (via API).

Data are normalised, log-transformed, and optionally reduced to a subset of marker genes (e.g., Yamanaka factors, senescence markers).

3.2 Therapy Generation

Based on the coherent bubble set, the system generates:

1. **OSKM cocktail** – expression levels of four Yamanaka factors (Oct4, Sox2, Klf4, c-Myc) are compared to a healthy reference; the required fold change is translated into a reprogramming protocol.
2. **Senolytics** – if senescence markers (p16, p21) are elevated, senolytic drugs (dasatinib+quercetin) are recommended.
3. **CRISPR targets** – differential expression between diseased and healthy bubbles identifies driver genes; CRISPR guides are proposed for top candidates.

3.3 Ethics Filter

The filter (implemented in `gra_core/ethics_box.py`) blocks:

- Therapies that would cause cell death (e.g., CRISPR knockouts of essential genes).
- Treatments predicted to increase tumorigenicity (based on oncogene activation).
- Any intervention that violates the “do no harm” principle as formalised in the Alan Law.

3.4 Integration with LLMs

An optional post-processing step sends the therapy proposal to an LLM (Ollama or OpenAI) for natural-language explanation. The GRA zeroing filter is applied to the LLM output to suppress hallucinations.

4 Experimental Demonstration: Breast Cancer Case

We applied GRA-CellTwin to a public breast cancer scRNA-seq dataset (GSE161529). The coherent bubble multiverse identified three sub-clusters: epithelial, mesenchymal, and an intermediate hybrid state. The therapy generator proposed:

- OSKM cocktail (Oct4, Sox2, Klf4, c-Myc) – to reprogram hybrid cells toward normal epithelium.
- Senolytics (dasatinib+quercetin) – to eliminate senescent cells expressing high p16.
- CRISPR guides to knock down MYC and TP53 (if mutated), and to activate known tumour suppressors.

All proposals passed the ethics filter. The notebook `demos/cell_simulator.ipynb` visualises the bubble trajectories and therapy outputs.

5 Conclusion and Future Work

GRA-CellTwin-Final demonstrates that GRA zeroing can create accurate, hallucination-free digital twins of single cells. By combining transcriptomic data with a multiverse foam, resonance filtering, and ethics-aware therapy generation, the system offers an open-source alternative to proprietary platforms (e.g., CZI Biohub). Future work includes:

- Integration with spatial transcriptomics for tissue-level twins.
- Real-time retraining of the foam as new data arrive.
- Automatic validation of generated therapies against external databases (e.g., CRISPick).

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